BUILD BRIEF · ALIANZA NIDO LANE · REFERENCE DOCUMENT

# NIDO Campo and the transparency dashboard

Two tools for the Pijao, Quindío rebuild: an offline-first Android app for documenting field visits, and a public dashboard that renders every peso and every minga from one ledger. Built to be adopted by other community projects.

> **Track 1, NIDO Campo**: field visit app, 8 builders, sections 1 to 13
>
> **Track 2, Transparency dashboard**: 3 to 4 builders, only after track 1 is staffed, sections 14 to 20
>
> **Partners**: Corporación Espejo de Agua · AHO! Wellness Tech · Esto es Quindío · NIDO | Hogares Vivos · La Semilla · Sancocho Club
>
> **License and ownership**: MIT, public repo, everything built belongs to the Pijao community
>
> **Companion documents**: Sprint pack (one card per squad) · Community explainer (EN/ES)

This is the reference. Builders start from the sprint pack; this document is where they look when a card is not enough. Prepared by AHO! Wellness Tech, October 20, 2026.

# 1. Why this exists

On August 10, 2026 a 7.4 earthquake hit Pijao, Quindío (Colombia); fires the next day destroyed the homes of 35 campesino families and left 43 Embera Chamí families without roofs. Alianza Nido (espejodeagua.org/nido) is rebuilding the countryside house by house with bahareque (guadua and earth) homes, weekly mingas (community work days) and a regenerative farm plan around each home. Phase 1 targets 40 homes, about 200 people and 40 hectares restored.

Today every field visit (a family census, a construction check, an agronomy visit, a minga, a delivery of materials) is captured on paper, WhatsApp and phone photos, then retyped by one person into the site's datos.json and photo manifests once a week. Evidence gets lost, donors see stale numbers, and the people doing the visits spend their evenings transcribing. The site is run by volunteers who will not pay for recurring cloud services, so the tool has to be cheap to run and simple to hand over.

The field team: Corporación Espejo de Agua (the local nonprofit), AHO! Wellness Tech (platform and coordination), La Semilla (construction, agroecology and birding know-how), the maestro de obra and his crew, and Saturday minga volunteers. Phones are mid-range to older Androids. Signal in the veredas is unreliable or absent; wifi is back in town.

# 2. What we are building

**NIDO Campo**: an Android app that lets a field worker pick the type of visit, answer a short questionnaire, record voice notes and take photos or videos with no signal, then uploads everything the next time the phone is online. On the server, uploads land in a review queue; an organizer approves them and the public site and the internal dashboard update.

1. **The app** (Android). Triage screen, questionnaire engine driven by a config file, capture (photo, video, voice), local queue, background sync.
2. **The sync server and review queue** (any stack the squad likes). Receives visits and media, deduplicates, stores, and shows a simple review page where an organizer approves, edits captions and publishes.
3. **The Gemini pipeline** (runs on the server, never in the field). Transcribes voice notes, fills questionnaire fields from the transcript, drafts captions and alt text in Spanish and English, flags photos that may show minors or identifiable faces for the consent review.

The app must be **generic**: visit types, questions, territory names and the server URL all come from one project-config.json. Change the file and the same APK serves another community project. Nido is the first project config, not the only one.

Who holds the phone: AHO! and Espejo de Agua coordinators in Pijao, La Semilla, the maestro de obra, Ancestral Organics in Filandia, and trusted minga volunteers. No user accounts: a per-territory invite code unlocks the app, and the device remembers the person's name.

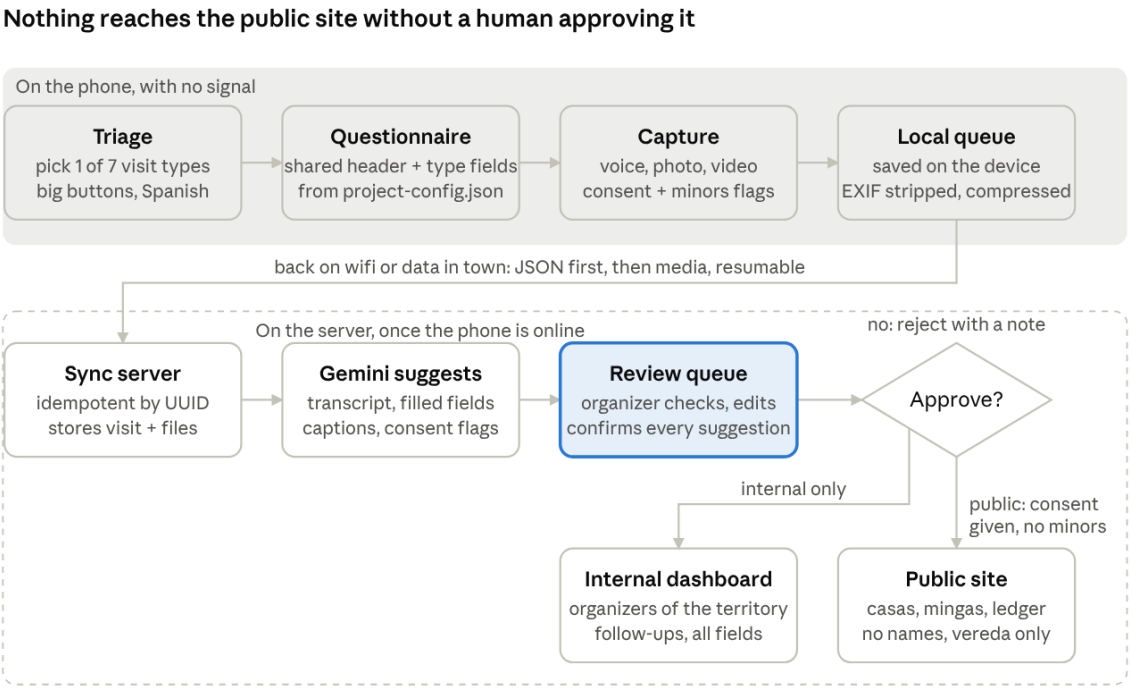
# 3. Hard constraints

These are not preferences. A build that breaks one of them cannot be handed to the community.

| Constraint | What it means for the build |
|---|---|
| Offline first | Every action works with airplane mode on: create visit, answer questions, record audio, take photos and video, review and edit past visits. Sync is a background job, never a blocker. |
| Older Android | Target Android 7.0 (API 24) and up. Test on a phone with 2 GB RAM. No feature may depend on a current flagship. |
| Spanish UI | All field-facing text in Colombian Spanish, short words, big tap targets. Field workers often have gloves, sun glare and one hand free. Strings live in one file so another language is a translation, not a rewrite. |
| Small storage, slow uploads | Compress photos on device (max 1600 px long edge), video to 720p, audio to AAC or Opus. A day's visits should fit in under 300 MB. Uploads resume per file after a drop. |

| Constraint | What it means for the build |
|---|---|
| No GPS in the data | The app never stores coordinates. Location is a vereda picked from a list, nothing finer. EXIF (GPS, device model, timestamps) is stripped from every photo before it leaves the phone. |
| Consent recorded | Every visit records who gave consent and for what (internal only, or publishable). Every photo or video carries two flags: identifiable people, minors present. Default is internal only. |
| Nothing public without review | The server never publishes an upload directly. Everything waits in a review queue; an organizer approves. |
| Free to run | No service with a recurring bill. Gemini on the free tier, hosting on the existing droplet or a free tier, MIT or Apache 2.0 license so Pijao and other projects own it. |
| Generic by config | Visit types, questions, veredas, invite codes, server URL and project name all come from project-config.json, not from code. |

# 4. How a visit flows




*The top row is the app and works with no signal; the lower rows run on the server once the phone is back online. Gemini only suggests, an organizer decides, and public is a separate approval from internal.*

# 5. Visit types and questionnaires

The first screen is the triage: seven big buttons, one per visit type. Picking one loads that type's questionnaire from project-config.json. Every type shares the same header and the same capture tools. Field labels are the Spanish the field team uses; keys in parentheses are the JSON keys. Types marked MVP are finished first on Saturday; the rest are config entries that reuse the same engine.

## Shared header (every visit)

| Field | Type | Notes |
|---|---|---|
| Fecha y hora (started_at) | auto | Device clock, ISO 8601 with offset |

| Field | Type | Notes |
|---|---|---|
| Tipo de visita (visit_type) | enum | From the triage |
| Municipio (municipio) | enum | Pijao, Filandia, … from config |
| Vereda (vereda) | enum + other | Finest geography allowed |
| Casa o finca (site_ref) | pick from list or new | Slug from config (e.g. golondrinas); new sites get a provisional name, never an address |
| Quién visita (author, companions) | text | Device owner prefilled; add others |
| Quién recibe (host_name) | text, optional | Only with consent; may be left blank |
| Consentimiento (consent) | object | given_by, scope: internal or publishable, verbal: true, recorded_in_audio |
| Notas de voz (audio[]) | media | Unlimited, each under 5 min, with a label chip |
| Fotos y videos (media[]) | media | Each with people_identifiable and minors_present set before saving |
| Nota libre (notes) | long text | Also fed by transcription |
| Seguimiento (follow_up) | yes/no + text + date | Drives the organizer's to-do list |

## A. Visita inicial / censo familiar (MVP)

First contact with a family or farm after the disaster. Counts, not identities.

- Personas en el hogar (household): adultos, niñas y niños, personas mayores, personas con movilidad reducida (numbers only)
- Estado de la vivienda (dwelling_state): habitable, dañada, destruida, en reconstrucción
- Dónde viven ahora (current_shelter): misma finca, con familia, albergue, arriendo, otro
- Servicios (services): agua, luz, saneamiento, señal celular (each: sí, parcial, no)
- Tierra y cultivos (land): hectáreas aprox, cultivos principales (café, plátano, frutales, huerta, pastos, otro), animales
- Pérdidas (losses): checklist (casa, cocina, cultivos, animales, herramientas, agua, camino) + text
- Necesidades urgentes (urgent_needs): ranked top 3 from list + text
- Quiere participar (participation): minga en su finca, aportar manos en otras, Red de Apoyo Ayni, Sanar y Sembrar
- Acuerdos y próximos pasos (agreements): text
- Never asked: cédula, dirección, ingresos, coordenadas, afiliación política o religiosa

## B. Visita técnica de obra (MVP)

A construction check on a house being built.

- Casa (site_ref): required, from the houses list
- Paso actual (current_step): enum from the house's pasos list (Demolición y limpieza, Excavación y zapatas, Vigas de cimentación, Contrapiso, Muros, Viga de amarre, Cubierta, Acabados y habitar)
- Avance del paso (step_progress): 0, 25, 50, 75, 100 %
- Qué se hizo desde la última visita (work_done): text or voice
- Materiales en sitio (materials_on_site): checklist from config + faltantes (text)
- Cuadrilla presente (crew_count): number
- Problemas o riesgos (issues): multi-select (humedad, cimiento, suministro, clima, seguridad, presupuesto) + text
- Gasto registrado hoy (expenses[]): rubro (enum from config), monto COP, comprobante (photo)
- Acta o entrega firmada (signed_record): photo of the signed sheet
- Próximo paso y fecha (next_step, next_date)

## C. Visita técnica agro / finca (MVP)

Agronomy or farm-design visit. Mirrors the "fichas" in Saber Vivo (espejodeagua.org/nido/saber-vivo): observe first, interpret later.

- Lote o sitio en la finca (plot_name): text
- Tipo de ficha (sheet_type): lote de cultivo, árbol, suelo, huerta, agua y laderas, cuidado de la casa
- Cultivo o especie (crop_or_species): enum + other (café, plátano, banano, aguacate, maíz, fríjol, zapallo, caña, guamo, nogal, chachafruto, guadua, otro)
- Etapa (stage): text or enum per crop
- Lo observado (observations): text or voice (symptoms, not diagnoses)
- Prueba de suelo (soil_test): tipo (lavado de manos, rollo, frasco, muro de prueba) + resultado
- Recomendación acordada (recommendation): text
- Insumos o semillas entregadas (inputs_delivered[]): item, cantidad, unidad
- Próxima visita (next_visit): date

## D. Minga o convite (MVP)

A community work day. Feeds the public mingas page and the Sancocho Club accounts.

- Minga (minga_ref): pick from scheduled list or new
- Propósito (purpose): text
- Asistentes (attendees_count): number; names only in attendee_names[] with consent
- Horas aprox (hours): number
- Lo hecho (work_done): checklist from config + text or voice
- La olla (meal): platos servidos, quién financió (Sancocho Club, aliado, familia, otro), costo COP si se conoce
- Herramienta compartida y semillas intercambiadas (tools_shared, seeds_exchanged): text
- Incidentes o seguridad (incidents): text
- Aprendizaje para compartir (lesson): one or two sentences for Saber Vivo en vivo

## E. Entrega de materiales o acta firmada

- Destino (site_ref): house or farm
- Ítems (items[]): ítem, cantidad, unidad, valor COP, fuente (donación, compra, aliado)
- Comprobante (receipt_media): photo of the invoice
- Acta firmada (signed_record): photo
- Recibido por (received_by): text, with consent

## F. Taller o acompañamiento (Sanar y Sembrar)

Group wellbeing, art and play sessions. Group level only, no notes about individuals.

- Tipo (session_type): taller, círculo, juego, teatro, otro
- Lugar (place): vereda or town venue, from list
- Participantes (participants): adultos, niñas y niños, personas mayores (counts only)
- Tema (topic): text
- Quién facilita (facilitators): organizations, not individuals
- Materiales usados (materials): text
- Observaciones generales del grupo (group_notes): text or voice
- Próxima sesión (next_session): date

### G. Nota libre

Header + voice, photo, video, text and tags (tags[]). For anything that does not fit, or when there is no time to fill a form in the field. Gemini can later propose a visit type and move fields across.

## 6. Data model and sync contract

The contract is the one thing all three squads must agree on by 11:30. Anything that speaks this contract can replace any other part later, which is what makes the tool portable to other projects.

### Visit record (campo/1)

One JSON document per visit. The app owns the id (UUID v4 generated on the phone), so the server can deduplicate retries. answers is a flat object whose keys come from the questionnaire config, so adding a question never changes the schema. Server-side fields (received_at, review.status, review.by, transcripts, published_to) are added by the server and never sent by the app.

```
{ "schema": "campo/1", "id": "6f1c2a4e-9b0d-4d3a-a6f2-0c5b8e7d1a22",
    "project": "nido-pijao",
  "visit_type": "obra", "started_at": "2026-10-31T09:12:00-05:00",
      "ended_at": "2026-10-31T10:05:00-05:00",
  "device_id": "a3f9...", "author": "Gustavo", "companions": ["maestro de obra"],
  "municipio": "Pijao", "vereda": "La Maizena", "site_ref": "golondrinas",
      "host_name": null,
  "consent": { "given_by": "familia", "scope": "internal", "verbal": true,
      "recorded_in_audio": false },
  "answers": { "current_step": "vigas-cimentacion", "step_progress": 50, "crew_count": 4 },
  "notes": "", "follow_up": { "needed": true, "what": "Falta cemento para contrapiso",
      "by": "2026-11-07" },
  "media": [
    { "id": "...", "kind": "photo", "file": "6f1c.../01.jpg", "bytes": 412000,
        "sha256": "...",
      "people_identifiable": false, "minors_present": false, "caption": "",
          "label": "vigas" },
    { "id": "...", "kind": "audio", "file": "6f1c.../a1.m4a", "duration_s": 84,
        "label": "materiales" } ],
  "app_version": "0.1.0", "config_version": "nido-pijao/3" }
```

### Sync endpoints

Plain HTTPS, JSON, no SDK. Auth is a bearer token derived from the invite code (Authorization: Bearer <project>:<invite-code>). The reference implementation can be 200 lines of Python, Node or Go; the squad picks.

| Method and path | Purpose | Rules |
|---|---|---|
| GET /api/campo/config | Download project-config.json | ETag so the app only refetches on change. The app ships with a bundled copy and works without ever calling this. |
| PUT /api/campo/visits/{id} | Upload or update one visit record | Idempotent. Same id twice returns 200 and the stored version. Body under 256 KB. |
| PUT /api/campo/visits/{id}/media/{media_id} | Upload one media file | Idempotent by sha256. Content-Range for resume, or the simpler rule: retry whole file, server discards duplicates. Max 200 MB. |
| GET /api/campo/visits/{id} | Confirm what the server holds | App marks the visit synced only when every media sha256 is confirmed. |

| Method and path | Purpose | Rules |
|---|---|---|
| GET /api/campo/sites | Houses, farms, mingas the app can pick from | Optional. Falls back to the list inside the config. |

Sync order on the phone: visit JSON first (small, so the organizer sees the visit exists), then media, smallest first. A visit shows one of four states: draft, queued, partial (JSON up, media pending), synced. Never delete local media until synced.

### Offline export bundle (fallback)

If a phone can never reach the server (or the server is down on demo day), the app exports a visit or a whole day as a .zip: visits/*.json plus media/ by visit id, and the person sends it by WhatsApp, USB or Bluetooth. The server has an Import bundle button on the review page that accepts the same zip. This is also how another project can use the app with no server at all.

### Review queue

One page, protected by the organizer login. Lists visits by received_at, newest first. For each: fields, transcript, media with the two consent flags, and three actions: **Approve for internal**, **Approve for public** (only if consent.scope is publishable and no media has minors_present: true), **Reject with a note**. Approving writes an event to the project's ledger (section 16) and, for Nido, produces the entries that today are typed by hand: a hito for the house, a photo manifest entry with alt, pie, consentimiento, menores, publicar, and the minga counts.

## 7. Privacy and consent, by design

The alliance's privacy notice (espejodeagua.org/nido/privacidad) makes promises to a community where naming a leader next to a location can put them at risk. The app must make the promises easy to keep and hard to break. Legal frame: Colombia's Ley 1581 de 2012 (habeas data); data controller is Corporación Espejo de Agua.

1. **Geography stops at the vereda.** No GPS permission requested. No coordinates in any record. Vereda is a dropdown. EXIF stripped on device before upload. Test: upload a photo, download it from the server, run exiftool: nothing but dimensions.
2. **No identity documents, no addresses.** The questionnaires never ask for cédula, dirección, income, religion or politics. The config validator rejects fields with those names.
3. **Consent is a required step, not a checkbox at the end.** Before the first photo or voice note, the app asks who gave consent and for what (internal or publishable). Default internal; switching to publishable takes a deliberate second tap. The field worker can optionally record the verbal consent as the first voice note.
4. **Minors are never identifiable.** Each photo or video gets two flags before it is saved. A photo flagged minors_present can never be approved for public, whatever the consent says. Gemini pre-fills the flags on the server; a human confirms.
5. **Counts, not names.** Household and workshop sections take numbers. Names are optional, only with consent, and never appear next to a vereda in anything public.
6. **Nothing public before it happens.** The server publishes activities after their ended_at, never a planned visit with a location.
7. **Data minimization on the server.** Store the device id, not the IP. Keep raw audio for 90 days after transcription, then delete unless pinned. The review page shows who approved what.
8. **Right to withdraw in 24 hours.** Every published item carries the visit id, so an organizer can find and unpublish it in one click. Rejected or withdrawn visits stay internal, not deleted, unless the person asks.
9. **Who sees what.** Internal: organizers of that territory only (Pijao: AHO! and Espejo de Agua; Filandia: Ancestral Organics), matched by the invite code's territory. Public: only approved, with the public-safe subset of fields.

## 8. Tech choice: native APK first, PWA as the fallback

The decision is made at 10:45 by one rule: **count the people in the app squad who arrive with `flutter doctor` passing (Android toolchain green) and a physical Android phone with USB debugging on. Two or more: build native. Fewer: build**

**the PWA and wrap it at the end.** Do not spend build time installing Android Studio; it is a 30 to 60 minute download that kills a 4.5 hour hack.

| Option | When | Stack | Why it fits | Watch out |
|---|---|---|---|---|
| Native APK (recommended) | 2+ people with Flutter ready | Flutter, drift or sqflite, image_picker, record, connectivity_plus, dio, flutter_image_compress | One codebase, real APK, strong offline packages, AI assistants write Flutter fluently, minSdk 24 | Release signing takes 10 min: do it at 11:00. A debug APK is fine for the demo. |
| Native APK, Kotlin | The squad is Android developers already | Kotlin, Jetpack Compose, Room, CameraX, MediaRecorder, WorkManager | Best camera and background behaviour on old phones | Slower to iterate on forms; only if the whole squad lives in Android Studio |
| PWA (fallback) | Fewer than 2 Flutter-ready laptops | Vite + React or vanilla, vite-plugin-pwa, Dexie, MediaRecorder, input capture, browser-image-compression | Runs on any laptop in minutes, installs to the home screen from Chrome, offline via service worker | No background upload. Test on a real phone over https (ngrok or cloudflared), not localhost. |
| PWA wrapped as APK | PWA done by 14:30 and one laptop has the Android SDK | Capacitor or Bubblewrap (TWA) | An installable APK from the same code in about 20 minutes | Capacitor camera and filesystem plugins improve reliability; add only if time allows |

Not recommended for Saturday: React Native with Expo (cloud builds queue for 15 to 40 minutes, local builds still need the Android SDK) and anything that needs an app store.

### Architecture inside the app, whichever stack

- **Config-driven forms.** One renderer turns project-config.json question definitions (text, number, enum, multi, date, counts, items[], media) into screens. No per-visit-type UI code.
- **Local store first.** Every save writes to the local database and the media folder. The sync worker reads from the store; the UI never waits on the network.
- **Sync worker.** Runs on app open, on connectivity change, and on Sincronizar. Walks queued and partial visits, uploads JSON then media, confirms hashes, marks synced. Exponential backoff, one upload at a time on mobile data.
- **Big buttons, few screens.** Triage, header, questions, capture, review, done. Nothing deeper than three taps. 18 px fonts, 48 dp tap targets, high contrast for sunlight.
- **Compression on save.** Photos 1600 px JPEG q80, video 720p and 90 s per clip, audio AAC 48 kbps mono or Opus 24 kbps.

Repository: one repo, app/, server/, ai/, config/nido-pijao/, fixtures/, docs/. MIT license. README with a 5-line quick start per folder. The alliance forks it after the event; the hackathon repo stays public.

## 9. Where Gemini fits (free tier only)

Gemini runs on the server at sync time, never on the phone in the field. Use a Gemini Flash model through a Google AI Studio API key on the free tier, or the MLH credits for the day. Every Gemini output is a **suggestion** that a human confirms on the review page; nothing Gemini writes goes public on its own. If the key or quota is missing, the pipeline skips and the visit still lands in the queue.

| Job | Input | Output | Notes |
|---|---|---|---|
| Transcribe voice notes | Audio file, Spanish | Transcript in transcripts[] | Prompt with the vereda and crop names from the config so proper nouns come out right |

| Job | Input | Output | Notes |
|---|---|---|---|
| Fill the questionnaire | Transcript + that type's question definitions | JSON matching answers keys, confidence per field | Structured output. Only fill empty fields; never overwrite what the field worker typed. Highlight filled fields on the review page. |
| Caption and alt text | Photo or video keyframe | alt_es, alt_en, pie_es, pie_en | Same style as the site's manifests: one plain sentence, no names |
| Consent flags | Photo or keyframe | people_identifiable, minors_present, faces_count | Pre-fill only. A true suggestion locks the public option until a human clears it. Err toward true. |
| Classify free notes | Nota libre text and transcript | Suggested visit_type and moved fields | For type G only |
| Weekly summary | Approved visits of the week | Two short paragraphs, ES and EN | Shared with the dashboard track; section 18 |

Budget: at most 1 transcription call per audio, 1 extraction call per visit, 1 vision call per media file. A 20-visit week is under 200 calls, far inside the free tier. Optional: Android's built-in SpeechRecognizer with the Spanish offline pack gives a rough on-device draft; the server transcript replaces it.

## 10. Team plan and timeline

**Eight builders finish the MVP in one day; six is the floor.** Sizing assumes everyone codes with an AI assistant: a config-driven form screen is a 20 to 30 minute task, a 200-line sync server is 45 minutes, a review page is an hour. The scarce resource is not typing, it is agreement: lock the contract early, keep squads to 2 or 3 so nobody waits on a merge, and give every squad a fixture to work against before the real thing exists.

| Squad | People | Owns | Done when |
|---|---|---|---|
| A1 App: triage and forms | 2 | Triage screen, config loader, questionnaire renderer, header, visit list and edit | Types D and B fill end to end from config with airplane mode on |
| A2 App: capture and sync | 2 | Photo, video, voice with consent flags, compression, EXIF strip, local store, sync worker, export zip | A visit with 3 photos and 1 voice note syncs after airplane mode is turned off |
| B Server and review | 2 | Contract endpoints, storage, dedup, review page with three actions, bundle import | Review page lists a synced visit, plays the audio, Approve writes a ledger event |
| C Gemini pipeline | 1 to 2 | Transcription, extraction with schema, captions and flags, hooked to upload | Transcript and suggested fields appear on the review page within a minute of sync |
| Integration lead | 1 | Contract, fixtures, demo phone, demo script, time calls | Demo rehearsed twice before 15:30 |

With 10 to 12 builders in the lane, staff this first and send the rest to the dashboard track. With 15, both tracks run comfortably and squad C can take the weekly summary too.

| Time | Everyone | Notes |
|---|---|---|
| 10:30 | Lane briefing: 5-minute pitch, the privacy rules, the demo we want at 15:45 | 15 minutes, no more |

| Time | Everyone | Notes |
|---|---|---|
| 10:45 | Form squads, run the Flutter-or-PWA rule, clone the repo with the contract and config | Integration lead publishes fixtures: a sample visit JSON, a mock server, media |
| 11:00 to 11:30 | Contract lock. All squads read section 6, raise changes, freeze campo/1 | After this, schema changes need the integration lead's OK |
| 11:30 to 13:00 | Vertical slice. A1 renders type D from config. A2 saves a visit and posts the JSON to B. B stores and lists it. C transcribes a sample m4a | Goal at 13:00: one minga visit created offline on a phone, synced, visible on the review page. No media yet. |
| 13:00 to 13:30 | Lunch, at the desk if the slice is not there yet | |
| 13:30 to 14:45 | Media and intelligence. A2 adds photo, voice, flags, compression, resume. B adds media endpoint and approve actions. C hooks transcription and extraction. A1 adds types A, B, C and the export zip | Goal at 14:45: full visit with media synced, transcript shown, Approve for public blocked by a minors_present flag |
| 14:45 to 15:15 | Polish and build. Debug APK or installed PWA on the demo phone. README. Config for a second fake project | Integration lead runs the demo once with airplane mode |
| 15:15 to 15:30 | Second rehearsal, submit | Repo link, 3-minute demo, one slide with the flow diagram |
| 15:45 | Demo | |

Rules that save an hour each: one repo and a main branch everyone pushes to with small commits; no login system beyond the invite code; no custom design system; no tests except one script that posts the fixture visit to the server; mock what you are waiting on, then swap.

## 11. Definition of done and demo script

Done means all of these pass on a physical Android phone, not an emulator:

- ☐ Airplane mode on: open app, pick Minga, fill the header and 4 questions, record a 20 s voice note, take 2 photos (one flagged minors_present), save. App shows queued.
- ☐ Still offline: reopen the visit, edit a field, add a third photo. Kill the app, reopen, the visit is intact.
- ☐ Airplane mode off: within 60 s the visit goes partial then synced without the user doing anything (or after one tap on Sincronizar for the PWA).
- ☐ Review page lists the visit with fields, transcript, 3 photos with flags. Approve for public is disabled while a photo has minors_present. Unflag it, approve, a ledger event is written.
- ☐ Downloaded photo has no EXIF location or device data.
- ☐ Swap config/nido-pijao/project-config.json for config/demo-other-project/ and the triage shows different visit types with no code change.
- ☐ Export zip from the phone imports on the review page.
- ☐ README quick start works on a fresh laptop for app/, server/ and ai/.

### The 3-minute demo

1. 20 s, the problem: a photo of the paper sheets and the WhatsApp threads. "Every visit is retyped once a week by one volunteer."
2. 60 s, the field: phone in airplane mode, pick Minga, speak the voice note in Spanish, snap two photos, flag one as showing a child, save. Show the queue.
3. 40 s, the sync: wifi on, visit turns synced. Switch to the review page: the visit, the transcript, the suggested fields highlighted.
4. 30 s, the guardrail: try Approve for public, blocked by the minor flag. Approve for internal instead. Show the ledger event and, if the dashboard made it, the counter moving.

5. 20 s, the point: swap the config, show the triage change. "Any community project can use this with a JSON file. MIT licensed, and Pijao owns it tonight."

## 12. Stretch goals

- **Organizer to-do list** from follow_up fields, grouped by site, with a done button.
- **Per-site history** in the app: open a house or farm and see past visits, last step, open follow-ups, offline.
- **Scheduled mingas list** from GET /api/campo/sites so the Minga type pre-fills purpose and site.
- **Ledger adapter for Nido**: on approve, write the hito, pasoActual and photo manifest entries directly in the site's datos.json and manifiesto.json format.
- **Expense capture with receipt OCR**: Gemini reads the invoice photo in type E and pre-fills items and COP amounts.
- **Peer sync**: phones share visits over hotspot or Bluetooth so one phone with data uploads for the crew.
- **Second language pack**: an en strings file, to prove the UI is translatable.
- **Low-end mode**: smaller photos and no video when free storage is under 500 MB.

## 13. Before Saturday: setup checklist

Send this to registered builders on Thursday. Anyone who arrives with it done saves the whole lane an hour.

- ☐ A laptop with git, Node 20+ and Python 3.11+ installed.
- ☐ Your AI coding assistant signed in and working (Claude Code, Gemini CLI, Cursor, Copilot, whichever you use).
- ☐ App squad: Flutter stable with flutter doctor showing the Android toolchain green, or Android Studio with SDK 34 and a device connected. Allow 45 minutes and a good connection; it will not happen at the venue.
- ☐ An Android phone (yours or a borrowed older one) with Developer options and USB debugging on, and a USB cable that carries data.
- ☐ A Google AI Studio account and a Gemini API key on the free tier.
- ☐ A GitHub account, so you can be added to the repo in the first 10 minutes.
- ☐ Read the sprint pack and open espejodeagua.org/nido once, including Casas and Mingas, so the vocabulary (minga, vereda, bahareque, nido, paso) is familiar.
- ☐ Optional: an older Android phone you can leave with the project. The team in Pijao will use it.

On the day the integration lead brings: the demo phone, the repo with config/nido-pijao/project-config.json and fixtures already in it, a sample m4a voice note in Colombian Spanish, 10 sample field photos with signed consent, a hotspot in case the venue wifi blocks ports, and a printed copy of the privacy rules.

TRACK 2 · ONLY IF TRACK 1 IS STAFFED

# 14. Transparency dashboard: why

The alliance promised donors that every peso can be traced to a house and every pot of sancocho to a minga. Today that promise is kept by hand: one volunteer edits nido/data/datos.json (fund raised, goal, houses, their steps, budget lines, milestones, labour quotes) and the Transparencia and Las casas pages render from it. The counters on the home page carry a "cifras actualizadas" date that is often a week or more old, and the Sancocho Club accounts live in a separate spreadsheet.

Every small relief effort has the same problem: numbers scattered across a GoFundMe page, a spreadsheet, WhatsApp receipts and a website. A generic, read-only dashboard that renders from one ledger file solves it for Nido and for the next project.

# 15. What we are building

**A public transparency dashboard that renders from one JSON ledger**, embeddable in any site with one <script> tag or an iframe, plus the small ledger service behind it. Donors see where money and work stand today; organizers update by appending events, not by editing a page.

1. **The ledger** (transparency/1): an append-only list of events (donation received, expense paid, step completed, minga held, delivery signed) plus a snapshot the dashboard reads. Events come from the field app's review queue, from a CSV import of the GoFundMe export, or from a tiny admin form.
2. **The dashboard**: a static web page that fetches the snapshot and draws it. Fund progress, per-house progress and budget, mingas and meals, last updated, and a verifiable trail (each number links to the events behind it). Spanish first, English toggle.
3. **The weekly report**: Gemini turns the week's events into two plain paragraphs, ES and EN, that an organizer approves and the dashboard shows under "Informe semanal".

Audiences, in order: donors in the US and Colombia checking that their gift moved; the alliance's partners reporting to their own networks; the community in Pijao seeing their work counted; and the next nonprofit that forks the repo.

# 16. Data contract: the transparency ledger

Two files, both JSON, both public except where marked. The ledger is the truth; the snapshot is derived from it by a script and is what the dashboard loads.

### Events (ledger.jsonl, one object per line, append only)

```
{"id":"evt_01J","at":"2026-09-19","type":"step_completed","site":"golondrinas","step":"demoli
    cion",
  "source":"campo:6f1c2a4e","by":"review"}
{"id":"evt_01K","at":"2026-09-19","type":"expense","site":"golondrinas","category":"transport
    e-mano-obra",
  "amount":1800000,"currency":"COP","status":"paid","receipt":"private"}
{"id":"evt_01M","at":"2026-10-01","type":"donation","channel":"gofundme","amount":1982,"curre
    ncy":"USD",
  "fund":"techo-camino-semilla","donors":41}
{"id":"evt_01N","at":"2026-10-03","type":"minga","site":"sendero-arenales","attendees":14,"me
    als":18,
  "meal_funder":"sancocho-club","source":"campo:9a77"}
{"id":"evt_01P","at":"2026-10-03","type":"delivery","site":"golondrinas",
  "items":[{"item":"cemento","qty":20,"unit":"bulto"}],"signed":true}
```

Event types for v1: donation, expense, quote, step_completed, step_progress, minga, delivery, milestone, correction (points at an earlier event id, never edits it). COP amounts are integer pesos; USD amounts are dollars; the snapshot carries the trm (exchange rate) used and its date.

### Snapshot (transparency.json, regenerated on every append)

```
{ "schema": "transparency/1",
  "project": { "name": "Alianza Nido", "org": "Corporación Espejo de Agua", "lang": ["es",
      "en"] },
  "updated_at": "2026-10-24T20:10:00Z", "trm": { "value": 3116, "as_of": "2026-10-03" },
  "funds": [ { "id": "techo-camino-semilla", "channel": "gofundme", "raised": 1982,
      "goal": 25000,
               "phase1_goal": 8000, "currency": "USD", "donors": 41 } ],
  "sites": [ { "id": "golondrinas", "name": "Casa 1 · Finca Las Golondrinas",
      "municipio": "Pijao",
               "status": "en-obra", "steps":
                   ["demolicion","excavacion","vigas","contrapiso","muros",
               "amarre","cubierta","habitar"], "current_step": 1, "progress_pct": 50,
               "budget": [ { "category": "materiales", "spent": 6000000, "committed": 0,
                   "needed": 0 },
                           { "category": "mano-de-obra", "spent": 1800000, "committed": 0,
                               "needed": 11000000 } ],
               "milestones": [ { "at": "2026-09-19", "text_es": "Minga de demolición",
                                 "text_en": "Demolition minga", "evidence": 3 } ] } ],
  "mingas": { "count": 2, "attendees": 27, "meals": 36,
      "by_funder": { "sancocho-club": 36 } },
  "deliveries": { "count": 1, "signed": 1 },
  "weekly_report": { "week": "2026-W43", "es": "...", "en": "...",
      "approved_by": "review" } }
```

Rules: no names of people anywhere in either file; sites are identified by slug and municipio only; receipts are stored privately and referenced as private (the dashboard shows "comprobante archivado", not the image); every snapshot number must be reproducible from the events by make snapshot. A one-off import script turns the current datos.json into the first ledger so Nido starts with real history on day one.

## 17. Views to build

One static page, no framework required (plain HTML, CSS and a small JS module is fine; React or Svelte if the squad prefers). Loads transparency.json, renders in under a second on a phone, works as an iframe or a script embed on espejodeagua.org.

| View | Shows | Source |
|---|---|---|
| Fund strip (top) | Raised vs goal and vs Phase 1 goal, donor count, channel, USD with COP equivalent at the stated TRM, updated_at | funds[], trm |
| Houses | One card per site: name, municipio, status, 8-step bar with current step and percent, budget by category (spent, committed, needed), latest milestone with evidence count | sites[] |
| House detail | Full step list with dates, budget table, milestone timeline, approved photos from the site's public manifest, "two ways to finish this house" (money, hands or materials) | sites[].milestones, budget |
| Mingas and the pot | Mingas held, people, meals served, meals by funder (Sancocho Club first), last minga date | mingas |
| Deliveries | Count, signed count, last delivery | deliveries |
| Weekly report | Two paragraphs ES and EN with the week label and who approved | weekly_report |

| View | Shows | Source |
|---|---|---|
| Trail | Every number is a link that opens the events behind it (filtered view of ledger.jsonl), so a donor can audit without asking | ledger.jsonl |

Design rules: Spanish default with an EN toggle (strings in one file); no names of people; numbers formatted by locale ($14.855.460 for COP, $1,982 for USD); the Nido palette (terracotta C8582B, moss green 2E4A36, cream F3EEE3) and type (Fraunces, Plus Jakarta Sans) if the squad has time, defaults otherwise; charts only where there are 5+ values. Must look right at 360 px wide.

## 18. Where Gemini fits

One job, done well: the weekly report. Input is the week's events plus the snapshot deltas. Output is two paragraphs, Spanish first in a Colombian register, then English, under 120 words each, no names, every number traceable to an event. Use structured output so the result is {"es": "...", "en": "...", "numbers_used": [...]} and the script can check that every number in the text exists in the events. An organizer approves the report; unapproved reports never show. Free tier, one call per week.

## 19. Team size and timeline

**Three builders finish it in one day; four is comfortable; two can ship the fund strip and houses view but not the trail or the report.** With AI-assisted coding, a static dashboard from a fixed JSON is a 90-minute task, the snapshot script is 45 minutes, and the ledger import from datos.json is 30 minutes; the day's risk is scope creep in the visuals, not code volume.

| Squad | People | Owns | Done when |
|---|---|---|---|
| D1 Ledger and snapshot | 1 | ledger.jsonl writer with validation, make snapshot, the datos.json import, a tiny append endpoint, the Approve hook | Importing today's datos.json yields a snapshot whose numbers match the live site |
| D2 Dashboard page | 2 | All views, locale formatting, EN toggle, embed mode, 360 px layout | Renders the real Nido snapshot and a fake second project with no code change |
| D3 Weekly report (optional) | 1 | Gemini call, number check, approve step, display | A report for the week of Oct 19 generates from imported events and shows after approval |

| Time | Plan |
|---|---|
| 10:45 | Form squad, repo folder dashboard/, read section 16 with squad B so Approve writes the same event shape |
| 11:00 to 11:30 | Freeze transparency/1. D1 hand-writes the first snapshot from datos.json as a fixture so D2 starts immediately |
| 11:30 to 13:00 | D2: fund strip and houses cards on the fixture. D1: import script and snapshot generator. D3: prompt and schema, first report from fake events |
| 13:30 to 14:45 | D2: house detail, mingas, trail, EN toggle, embed. D1: append endpoint and the Approve hook, wired to squad B. D3: approval step and display |
| 14:45 to 15:15 | Second project config, mobile check at 360 px, README, rehearse the 90-second demo |
| 15:30 | Submit. Demo: open the live snapshot, click a number to see its events, swap to the second project |

# 20. Definition of done, demo and stretch

- ☐ python import_datos.py nido/data/datos.json produces a ledger whose snapshot shows USD 1,982 raised of 25,000, Casa 1 on step 2 of 8, materials COP 6,000,000 spent, labour COP 1,800,000 spent and COP 10 to 12 million needed, matching the live site today.
- ☐ Appending a minga event and re-running the snapshot moves the mingas and meals counters.
- ☐ The dashboard renders that snapshot on a phone at 360 px with no horizontal scroll, in Spanish, with the EN toggle working.
- ☐ Clicking the raised amount opens the list of donation events behind it.
- ☐ Swapping to config/demo-other-project/transparency.json renders a different project with no code change.
- ☐ Embed snippet works inside a plain HTML page served from another folder.
- ☐ If D3 was staffed: a weekly report generates, fails validation when a number is invented, and shows only after approval.

Demo (90 seconds): the real site's counters frozen at a date, then the dashboard live with today's numbers; click one number, show the events; append a minga event from the field app's Approve and watch the counter move; swap project, same page, different nonprofit.

### Stretch

- GoFundMe CSV import for donations (count, amount, date, no donor names); Nequi or Vaki import for COP donations.
- datos.json writer: regenerate the current site's file from the snapshot so existing pages keep working until replaced by the embed.
- Per-partner view: what Sancocho Club, Esto es Quindío or NIDO | Hogares Vivos contributed, for their own reporting.
- Printable one-page PDF of the snapshot for the weekly WhatsApp report.
- Signed-record gallery: approved photos of actas firmadas, with consent, as the audit trail for deliveries.

# Links

- Event: luma.com/h4h-pdx, Hack for Humanity Portland, The AI Collective with Google and MLH
- Project: espejodeagua.org/nido · /nido/casas · /nido/mingas · /nido/transparencia · /nido/saber-vivo · /nido/privacidad
- Organization: Corporación Espejo de Agua, Pijao, Quindío; data controller under Ley 1581 de 2012
- Existing site stack, for squad B if they want to match it: static HTML on an Ubuntu droplet behind nginx, a stdlib Python form API under /api/, hand-edited nido/data/datos.json, per-event manifiesto.json photo manifests generated by deploy/fotos.py (strips EXIF, writes consent fields). Repo access on request.
- Lane contacts: Gustavo Díaz (AHO!, platform) · Yemayá Hall-Ruiz (AHO!, The AI Collective Portland)